THE SELF MAP GIVEN BY A SECTION IS HOMOTOPIC TO THE IDENTITY ON THE ASSOCIATED SIMPLICIAL OBJECT

THE STACKS PROJECT AUTHORS

Editorial note. The counted statement is the author's own lemma environment labelled lemma-fibre-products-simplicial-object-w-section (source lines 4873-4884); the proof environment begins on the second line after it and is its own proof, closing at source line 4906.

Editorial note. The proof explicitly leaves the verification of the relations to the reader while pointing at the relations-homotopy lemma; the retained card reproduces the author's text as written and does not supply the omitted verification, which is recorded in the pairing note rather than silently completed.

Editorial note. The retained preamble is the chapter's own minimal preamble (sources/stacks-simplicial/preamble.tex, taken with its real line count), a verbatim excerpt of the native Xy-pic setup, the native theorem declarations with the mathematical macros, and the project's own no-op reference environments; no macro was invented or redefined.

Editorial scope. One original result of the Stacks Project's Simplicial Methods chapter, counted once: when f has a section, the degreewise self map (s f)^{n+1} on the fibre-product simplicial object is homotopic to the identity, so the object is homotopy equivalent to the constant one.

Allowed standard prerequisites (editorial).

- The fibre-product simplicial object. The simplicial object U with U_n the (n+1)-fold fibre product of Y over X, built naively from f : Y to X when C has fibre products (the author's example-fibre-products-simplicial-object, retained as context).. It is the object of the statement (source lines 4877-4880).
- Sections and the maps s f. A section s of f with f s = id, hence an endomorphism s f of Y whose iterates appear in each degree.. It produces the self map of the statement and the g^1 used in the homotopy (source lines 4876, 4887).
- The relations characterising a homotopy of simplicial morphisms. A family h_{n,i} defines a homotopy if and only if it satisfies the five relations with the face and degeneracy maps (the author's lemma-relations-homotopy and remark-homotopy-better, retained as context).. It is the criterion the written-down family is claimed to satisfy (source lines 4901-4903).
- The functor of points. Describing a morphism of iterated fibre products by its effect on generalised points.. It is the language in which the homotopy is first described (source lines 4888-4897).

Date: stacks-project:simplicial.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Example 0.1. Suppose that $Y \to X$ is a morphism of $\mathcal{C}$ such that all the fibred products $Y \times_X Y \times_X \ldots \times_X Y$ exist. Then we set U_n equal to the $(n+1)$-fold fibre product, and we let $\varphi : [n] \to [m]$ correspond to the map (on "coordinates") $(y_0, \ldots, y_m) \mapsto (y_{\varphi(0)}, \ldots, y_{\varphi(n)})$. In other words, the map $U_0 = Y \to U_1 = Y \times_X Y$ is the diagonal map. The two maps $U_1 = Y \times_X Y \to U_0 = Y$ are the projection maps.

Lemma 0.2. *In the situation above, we have the following relations:*

1. *We have $h_{n,0} = b_n$ and $h_{n,n+1} = a_n$.*
2. *We have $d^n_j \circ h_{n,i} = h_{n-1,i-1} \circ d^n_j$ for $i > j$.*
3. *We have $d^n_j \circ h_{n,i} = h_{n-1,i} \circ d^n_j$ for $i \leq j$.*
4. *We have $s^n_j \circ h_{n,i} = h_{n+1,i+1} \circ s^n_j$ for $i > j$.*
5. *We have $s^n_j \circ h_{n,i} = h_{n+1,i} \circ s^n_j$ for $i \leq j$.*

Conversely, given a system of maps $h_{n,i}$ satisfying the properties listed above, then these define a morphism h which is a homotopy from a to b.

Proof. Omitted. You can prove the last statement using the fact, see Lemma 0.4 that to give a morphism of simplicial objects is the same as giving a sequence of morphisms h_n commuting with all d^n_j and s^n_j. □

Remark 0.3. Let $\mathcal{C}$ be any category (no assumptions whatsoever). Let U and V be simplicial objects of $\mathcal{C}$. Let $a, b : U \to V$ be morphisms of simplicial objects of $\mathcal{C}$. A *homotopy from a to b* is given by morphisms[1] $h_{n,i} : U_n \to V_n$, for $n \geq 0$, $i = 0, \ldots, n+1$ satisfying the relations of Lemma 0.2. As in Definition 0.5 we say the morphisms a and b are *homotopic* if there exists a sequence of morphisms $a = a_0, a_1, \ldots, a_n = b$ from U to V such that for each $i = 1, \ldots, n$ there either exists a homotopy from a_{i-1} to a_i or there exists a homotopy from a_i to a_{i-1}. Clearly, if $F : \mathcal{C} \to \mathcal{C}'$ is any functor and $\{h_{n,i}\}$ is a homotopy from a to b, then $\{F(h_{n,i})\}$ is a homotopy from $F(a)$ to $F(b)$. Similarly, if a and b are homotopic, then $F(a)$ and $F(b)$ are homotopic. Since the lemma says that the newer notion is the same as the old one in case finite coproduct exist, we deduce in particular that functors preserve the original notion whenever both categories have finite coproducts.

Lemma 0.4. *The category Δ is the universal category with objects $[n]$, $n \geq 0$ and morphisms δ^n_j and σ^n_j such that (a) every morphism is a composition of these morphisms, (b) the relations listed in Lemma 0.6 are satisfied, and (c) any relation among the morphisms is a consequence of those relations.*

Proof. Omitted. □

Definition 0.5. Let $\mathcal{C}$ be a category having finite coproducts. Suppose that U and V are two simplicial objects of $\mathcal{C}$. Let $a, b : U \to V$ be two morphisms.

(1) We say a morphism

$$h : U \times \Delta[1] \longrightarrow V$$

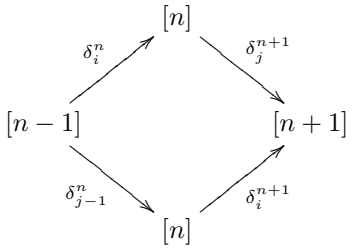
is a *homotopy from a to b* if $a = h \circ e_0$ and $b = h \circ e_1$.

[1] In the literature, often the maps $h_{n+1,i} \circ s_i : U_n \to V_{n+1}$ are used instead of the maps $h_{n,i}$. Of course the relations these maps satisfy are different from the ones in Lemma 0.2.

(2) We say the morphisms a and b are *homotopic* or are *in the same homotopy class* if there exists a sequence of morphisms $a = a_0, a_1, \ldots, a_n = b$ from U to V such that for each $i = 1, \ldots, n$ there either exists a homotopy from a_{i-1} to a_i or there exists a homotopy from a_i to a_{i-1}.

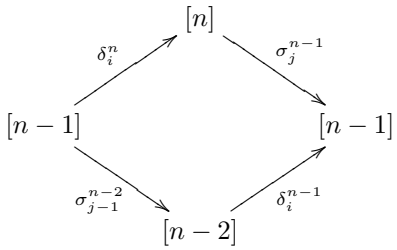
Lemma 0.6. *The morphisms δ^n_j and σ^n_j satisfy the following relations.*

(1) *If $0 \leq i < j \leq n+1$, then $\delta^{n+1}_j \circ \delta^n_i = \delta^{n+1}_i \circ \delta^n_{j-1}$. In other words the diagram*



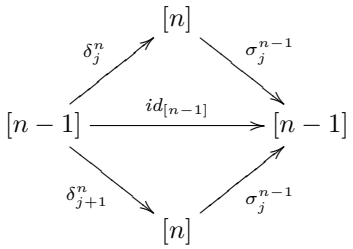
commutes.

(2) *If $0 \leq i < j \leq n-1$, then $\sigma^{n-1}_j \circ \delta^n_i = \delta^{n-1}_i \circ \sigma^{n-2}_{j-1}$. In other words the diagram*



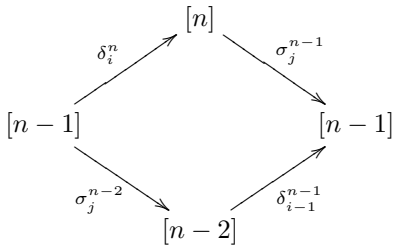
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(3) *If $0 \leq j \leq n-1$, then $\sigma^{n-1}_j \circ \delta^n_j = id_{[n-1]}$ and $\sigma^{n-1}_j \circ \delta^n_{j+1} = id_{[n-1]}$. In other words the diagram*



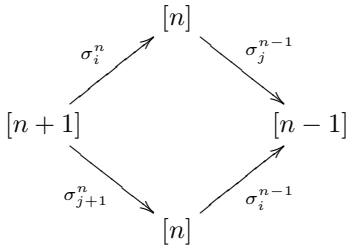
commutes.

(4) *If $0 < j+1 < i \leq n$, then $\sigma_j^{n-1} \circ \delta_i^n = \delta_{i-1}^{n-1} \circ \sigma_j^{n-2}$. In other words the diagram*



commutes.

(5) *If $0 \leq i \leq j \leq n-1$, then $\sigma_j^{n-1} \circ \sigma_i^n = \sigma_i^{n-1} \circ \sigma_{j+1}^n$. In other words the diagram*



commutes.

Proof. Omitted. □

Lemma 0.7. *Let $f : Y \to X$ be a morphism of a category $\mathcal{C}$ with fibre products. Assume f has a section s. Consider the simplicial object U constructed in Example 0.1 starting with f. The morphism $U \to U$ which in each degree is the self map $(s \circ f)^{n+1}$ of $Y \times_X \ldots \times_X Y$ given by $s \circ f$ on each factor is homotopic to the identity on U. In particular, U is homotopy equivalent to the constant simplicial object X.*

Proof. Set $g^0 = \mathrm{id}_Y$ and $g^1 = s \circ f$. We use the morphisms

$$\begin{aligned} Y \times_X \ldots \times_X Y \times \mathrm{Mor}([n], [1]) &\to Y \times_X \ldots \times_X Y \\ (y_0, \ldots, y_n) \times \alpha &\mapsto (g^{\alpha(0)}(y_0), \ldots, g^{\alpha(n)}(y_n)) \end{aligned}$$

where we use the functor of points point of view to define the maps. Another way to say this is to say that $h_{n,0} = \mathrm{id}$, $h_{n,n+1} = (s \circ f)^{n+1}$ and $h_{n,i} = \mathrm{id}_Y^{i+1} \times (s \circ f)^{n+1-i}$. We leave it to the reader to show that these satisfy the relations of Lemma 0.2. Hence they define the desired homotopy. See also Remark 0.3 which shows that we do not need to assume anything else on the category $\mathcal{C}$. □